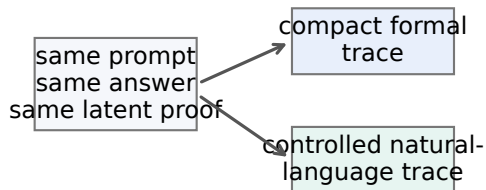


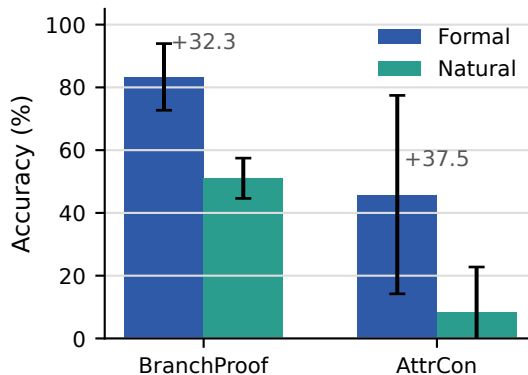
Formal traces improve extrapolative answers, but proof validity remains a separate target

A. Controlled substrate comparison

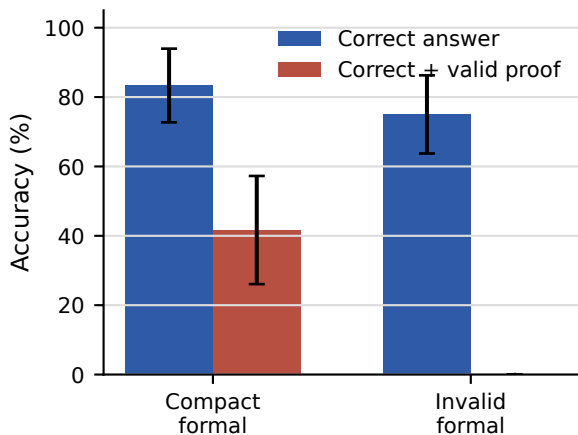


Only the trace substrate changes.

B. Depth-50 answer correctness



C. Correctness is not proof faithfulness



Formal depth-50 success

```
<proof> ... Uo ; ->E <answer>  
olive
```

Natural depth-50 failure

```
<think> premise chain ... (no  
answer tag before stop)
```

Invalid formal success

```
<proof> repeated R steps ...  
<answer> birch
```

Snippets are illustrative logged samples; metrics use 16 samples per prompt.